



TRAINING PROGRAMME

Enterprise Networking, Security & Automation (CCNA 3)

Connect, secure, manage and automate a multi-site enterprise network

42 HOURS

FOR Participants who completed CCNA 2 and are preparing the CCNA exam

PREPARES FOR CCNA (200-301 v1.1), course 3 of 3



Mohamed Aziz BEN HAHA

mohamedazizbenhaha.netlify.app

Overview

This programme is the third of three courses that prepare the Cisco CCNA 200-301 exam, aligned with Cisco Networking Academy Enterprise Networking, Security and Automation v7. Participants route between sites with single-area OSPFv2, filter traffic with access control lists, connect to the internet with NAT and PAT, and study WAN technologies, VPNs with IPsec, and QoS. They manage, design and troubleshoot the network, and automate it with REST APIs and Python. A timed practice exam covers the three courses, and the programme closes with a final project: a documented enterprise branch network.

At a glance

#	PHASE	FOCUS	HOURS
01	OSPF	Single-area OSPFv2 configuration, tuning and troubleshooting	6 h
02	Security, ACLs & NAT	Threats, access control lists, NAT and PAT	9 h
03	WAN, VPN & Network Management	WAN and SD-WAN, VPNs and IPsec, QoS, management protocols	6 h
04	Design, Automation & CCNA Review	Scalable design, troubleshooting, SDN, automation, practice exam	9 h
05	Final Project	An enterprise branch network, documented and tested	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		42 h

Key facts

AUDIENCE	Participants who completed CCNA 2 and are preparing the CCNA exam
PREREQUISITES	Switching, Routing & Wireless Essentials (CCNA 2), or equivalent: VLANs, inter-VLAN and static routing. These topics are not taught again; no prior Python needed
DURATION	42 h: 36 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs on simulated Cisco devices; one timed practice exam
LAB ENVIRONMENT	Runs locally: one laptop per participant with Cisco Packet Tracer (free with a Cisco Networking Academy account; its network controller serves the REST API labs) and Python 3; physical Cisco routers for selected labs when the organiser provides a rack; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase, a timed practice exam and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can build, secure and manage a multi-site network with OSPF, ACLs and NAT, read device data through a REST API with Python, and sit the CCNA 200-301 exam

PHASE 01

6 H

OSPF

Participants replace static routes with a dynamic routing protocol: single-area OSPFv2, its neighbours and link-state database, then DR and BDR election, cost tuning and default route propagation. Validated lab: three sites routed by OSPF, with a default route to the internet learned by every router.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Single-Area OSPFv2	OSPF concepts, configuration and verification	Lecture + Lab	3 h
1.2	OSPF in Practice	Multi-access networks, DR and BDR, cost, default route propagation, troubleshooting	Lecture + Lab	3 h

PHASE 02

9 H

Security, ACLs & NAT

Participants review the threats an enterprise network faces, then filter traffic with standard, extended and named ACLs placed where they belong, and connect private addressing to the internet with static NAT, dynamic NAT and PAT. Validated lab: a security policy written as ACLs, tested against allowed and denied flows, with users reaching the internet through PAT.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Network Security Concepts	Threats, attack types, security best practices	Lecture + Workshop	1.5 h
2.2	ACL Concepts	Wildcard masks, ACL types, placement	Lecture + Lab	1.5 h
2.3	ACL Configuration	Standard, extended and named IPv4 ACLs, securing VTY access	Lecture + Lab	3 h
2.4	NAT for IPv4	Static NAT, dynamic NAT, PAT	Lecture + Lab	3 h

PHASE 03

6 H

WAN, VPN & Network Management

Participants compare WAN technologies, see how site-to-site and remote-access VPNs protect traffic with IPsec and GRE, how QoS gives priority to sensitive traffic, then manage devices with CDP, LLDP, NTP, SNMP and syslog and back up configurations and IOS images. Validated lab: routers synchronised by NTP, sending syslog and SNMP data to a management server, with configurations backed up.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	WAN Concepts	WAN topologies and technologies, broadband, SD-WAN overview	Lecture + Workshop	1.5 h
3.2	VPN & IPsec Concepts	Site-to-site and remote-access VPNs, IPsec, GRE	Lecture + Lab	1.5 h
3.3	QoS Concepts	Traffic characteristics, queuing, QoS models and mechanisms	Lecture + Workshop	1.5 h
3.4	Network Management	CDP and LLDP, NTP, SNMP, syslog, configuration backup and IOS image management	Lecture + Lab	1.5 h

PHASE 04

9 H

Design, Automation & CCNA Review

Participants design hierarchical networks that scale and troubleshoot them with a method, see how virtualization, SDN and controllers such as Catalyst Center change network operations, and read device data through a REST API with a short Python script. A timed practice exam across the three CCNA courses closes the phase. Validated lab: a Python script that lists the devices known to the Packet Tracer network controller.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Network Design & Troubleshooting	Hierarchical and scalable design, troubleshooting methodology and tools	Lecture + Lab	3 h
4.2	Network Virtualization	Cloud, virtualization, SDN and controller-based networking	Lecture + Workshop	1.5 h
4.3	Network Automation	REST APIs, JSON and YAML, configuration management tools, Catalyst Center, AI in network operations	Lecture + Lab	1.5 h
4.4	CCNA 200-301 Review	Timed practice exam across the three courses and debrief	Lab	3 h

PHASE 05

6 H

Final Project

Each participant builds the network that connects a branch to its head office from a written brief, reusing the phase labs, then documents and tests it before presenting it to the group.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, then configure OSPF between sites, the ACL security policy and PAT to the internet	Project	3 h
5.2	Manage, Automate & Document	SNMP, syslog and NTP; a Python script that reads device data through a REST API; site-to-site VPN design; test plan and documentation	Project	1.5 h
5.3	Demo & Review	Test results and live demo; questions and feedback	Project	1.5 h

FINAL PROJECT

Enterprise Branch Network

A branch connected to its head office: OSPF between sites, an ACL security policy, PAT to the internet, a site-to-site VPN design, SNMP, syslog and NTP management, and a Python script that reads device data through a REST API; documented and tested.

STACK Cisco Packet Tracer and its network controller, Cisco IOS routers and switches, OSPFv2, ACLs, NAT, SNMP, syslog, NTP, REST API, Python

DELIVERABLE Packet Tracer file, Python script, design document, test results, live demo